



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 8 • STANDARD 6.DS.5

Display Data: Box Plots

Lesson 8-5 • Teacher Copy

Name: _____ **Date:** _____ **Class:** _____



TODAY'S OBJECTIVES

Content Objective: I can make and read a box plot to summarize a data set.

Language Objective: I can explain my box plot using the words box plot, median, quartile, and interquartile range.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Box Plot

Median

Quartile

Interquartile Range

Data distribution

Variability



Tap any word to see what it means and a picture.

- 1 A graph that shows data with a box and whiskers based on the five-number summary is a

- 2 The middle value of the data, shown as a line inside the box, is the

- 3 A value that divides the data into four equal parts is a .

- 4 The distance between the first and third quartiles, showing the middle 50% of data, is the

- 5 The way data values are spread out is the .

- 6 How spread out the data values are is the .

Reflect — Exit Ticket

A box plot has $Q1 = 20$ and $Q3 = 36$. What is the IQR and what does it represent?

- A. IQR = 16; it is the spread of the middle 50% of the data
- B. IQR = 56; it is the total of $Q1$ and $Q3$
- C. IQR = 28; it is the median between $Q1$ and $Q3$
- D. IQR = 36; it is the value of $Q3$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Box Plot
2. **Notes 2:** Median
3. **Notes 3:** Quartile
4. **Notes 4:** Interquartile Range
5. **Notes 5:** Data distribution
6. **Notes 6:** Variability
7. **Watch for this mistake:** *A common mistake in Display Data: Box Plots is skipping the key idea: "The box always holds the middle 50% of the data, and the line inside the box is the median." — always check your work against this rule before you submit.*
8. **Try It 1:** A. $Q1 = 12$, $Q3 = 28$ — *With 8 values the median splits the data into a lower half (6, 10, 14, 18) and an upper half (22, 26, 30, 34). $Q1$ is the median of the lower half: $(10 + 14) \div 2 = 12$. $Q3$ is the median of the upper half: $(26 + 30) \div 2 = 28$.*
9. **Try It 2:** A. Between 15 and 30 — *The box of a box plot stretches from $Q1$ to $Q3$, and that box always holds the middle 50% of the data. Here the box goes from $Q1 = 15$ to $Q3 = 30$.*
10. **Exit Ticket:** A. $IQR = 16$; it is the spread of the middle 50% of the data — $IQR = Q3 - Q1 = 36 - 20 = 16$. *The IQR tells you the range of the middle 50% of the data.*